

Hardware Components & System Assembly

PROGRAM	ASSESSMENT CODE	TOTAL TIME	STACK
CompTia A+	AP-A1	3 Hours	IT Support

GENERAL INSTRUCTIONS

Attempt all sections. Section B and C practical tasks must be performed on real desktop hardware where available; if physical hardware is not available, complete them using a PC-building/hardware simulator or clearly labeled written documentation and diagrams. Submit photos or screenshots of your practical work alongside your written answers. Label every response clearly with its section and task/question number. All work must be your own.

A

Section A — Concept Application

Answer all questions. Each question presents a real-world scenario followed by a concept-specific question.

S
1

SCENARIO

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

S
2

SCENARIO

You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

S
3**SCENARIO**

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

S
4**SCENARIO**

You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

S
5**SCENARIO**

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

S
6**SCENARIO**

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

B**Section B — Practical Coding Tasks**

Complete all tasks using the desktop PC components and tools provided. Each task targets one specific skill from the module. Follow the requirements listed under each task.

Task 1: Component Identification & Functions

Identify and document the functions of the core hardware components in a desktop system. (Foundational — direct identification of a single concept.)

- Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
- Note the make/model or key specification of each component where visible.
- Present the findings as a labeled component list or table.

Task 2: Motherboard Ports & Socket Identification

Identify the motherboard's physical layout, ports, and CPU socket type. (Applied – one concept examined in detail across multiple points.)

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).
- Identify all expansion slots (PCIe, M.2, etc.) and their purpose.
- Identify the CPU socket type and note which CPU families are compatible with it.
- Photograph or diagram the motherboard with labels.

Task 3: RAM & Storage Installation

Remove and reinstall a RAM module and a storage device. (Intermediate – two components handled together with correct procedure.)

- Safely remove the RAM module, noting ESD precautions taken.
- Reinstall the RAM module correctly and confirm it is seated properly.
- Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).
- Confirm both components are detected correctly after reinstallation.

Task 4: Full Desktop PC Assembly

Assemble a desktop PC from the provided components and document the process. (Integrative – combines all Section B skills into one build.)

- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.
- Follow correct installation sequence and ESD/safety precautions throughout.
- Document each step with labeled photos or diagrams.
- Power on the system and confirm it passes POST successfully.

C

Section C – Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Objective:

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

Your project must:

- Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.
- Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.
- Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.
- Produce a hardware–installation checklist of at least three steps to prevent similar failures in future deployments.
- Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

D

Section D — AI-Augmented Learning

STEP 1 • BUILD WITH AI

Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on-site.
- Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.

STEP 2 • TEST & DEBUG (WITHOUT AI)

Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself.

SUBMIT

All 3 items required

- 1 The exact prompt(s) you gave the AI tool.
- 2 The AI's original checklist and your corrected version.
- 3 A 3–4 line note explaining what you changed and why the AI's version needed it.